

Project: WellBeing360

Employee Wellness & Benefits Administration Platform

Domain: HR & Workforce

1. Introduction

WellBeing360 is a web-based employee wellness and benefits administration platform designed for enterprises and HR shared services teams to manage group health insurance enrolments, flexible benefits selection, wellness program management, Employee Assistance Program access, employee reward and recognition programs, and benefits utilisation analytics. The system enables employees to enrol in benefits plans, participate in wellness challenges, access EAP services, and redeem recognition awards; HR teams to configure benefit plans and manage open enrolment windows; and finance teams to track premium costs, reimbursement claims, and benefits spend budgets.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **Employee:** Enrols in benefits plans, submits wellness activity logs, accesses EAP, and redeems recognition points
- **HR Benefits Administrator:** Configures benefit plans, manages open enrolment windows, and resolves employee queries
- **Finance Executive:** Tracks premium costs, manages reimbursement claim approvals, and monitors benefits budget
- **Wellness Program Coordinator:** Designs and manages wellness challenges, tracks participation, and rewards completion
- **Recognition Manager:** Manages peer recognition awards, milestone rewards, and leaderboard programs
- **HR Admin:** Configures benefits catalog, plan eligibility rules, wellness templates, and user roles

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.

- **2.2 Benefits Plan Configuration & Catalog:** Manages group insurance, flexible benefits, and supplementary benefit plan definitions.
- **2.3 Employee Benefits Enrolment Management:** Handles open enrolment windows, plan selection, dependent registration, and coverage confirmation.
- **2.4 Wellness Program & Challenge Management:** Manages wellness initiatives, activity tracking, challenge leaderboards, and completion rewards.
- **2.5 Employee Assistance Program Management:** Manages EAP service access, counselling session bookings, and confidential usage tracking.
- **2.6 Recognition & Rewards Management:** Manages peer recognition programs, milestone awards, points accrual, and redemption catalog.
- **2.7 Benefits Analytics & Reporting:** Provides dashboards for enrolment coverage, wellness participation, claims spend, and recognition activity.
- **2.8 Notifications & Alerts:** Enrolment window alerts, wellness challenge reminders, EAP session confirmations, and recognition point updates.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- Employee and HR staff registration with role-based and benefit-plan-scoped access
- Confidential EAP usage tracking with privacy controls and full audit logging

Entities

- **User**
 - UserID
 - EmployeeID
 - Name

- Role
(Employee/HRBenefitsAdmin/Finance/WellnessCoordinator/RecognitionManager/Admin)
- Email
- Phone
- GradeID
- DepartmentID
- Status (Active/Inactive)
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - Module
 - Timestamp

4.2 Benefits Plan Configuration & Catalog Module

Features

- Configure group health insurance, life cover, and flexible benefit plans with eligibility tiers
- Define plan options, coverage limits, premium contribution splits, and effective dates

Entities

- **BenefitPlan**
 - PlanID
 - PlanName
 - PlanType
(GroupHealthInsurance/LifeCover/DentalVision/FlexibleBenefit/RetirementContribution/Commuter)
 - EligibilityGrade
 - EmployeeContribution
 - EmployerContribution
 - CoverageLimit

- EffectiveDate
- Status (Active/Discontinued)
- **FlexBenefitBucket**
 - BucketID
 - PlanID
 - BucketName (Medical/Childcare/Fitness/Education/Transport/Meal)
 - AnnualAllowance
 - CarryForwardAllowed
 - Status (Active/Inactive)

4.3 Employee Benefits Enrolment Management Module

Features

- Manage annual open enrolment windows with plan selection and dependent registration
- Track enrolment status, coverage activation, and manage mid-year life event updates

Entities

- **EnrolmentWindow**
 - WindowID
 - PlanYear
 - OpenDate
 - CloseDate
 - EligibleGrades
 - Status (Upcoming/Open/Closed)
- **BenefitEnrolment**
 - EnrolmentID
 - EmployeeID
 - PlanID
 - WindowID
 - CoverageOption

- DependentsIncluded
- EmployeeContributionAmount
- EffectiveDate
- Status (Draft/Submitted/Active/Lapsed/Cancelled)
- **Dependent**
 - DependentID
 - EmployeeID
 - Name
 - Relationship (Spouse/Child/Parent)
 - DateOfBirth
 - Status (Active/Removed)

4.4 Wellness Program & Challenge Management Module

Features

- Launch wellness challenges with activity categories, points, and participation targets
- Track employee activity submissions and compute leaderboard rankings

Entities

- **WellnessProgram**
 - ProgramID
 - Name
 - Theme (Fitness/Nutrition/MentalHealth/Preventive/WorkLifeBalance)
 - StartDate
 - EndDate
 - PointsOnOffer
 - TargetParticipation
 - Status (Upcoming/Active/Completed)
- **WellnessChallenge**
 - ChallengeID

- ProgramID
- ChallengeName
- ActivityType (Steps/Meditation/WaterIntake/SleepLog/NutritionTrack)
- DailyTarget
- Duration
- PointsPerCompletion
- Status (Active/Completed)
- **ActivityLog**
 - LogID
 - ChallengeID
 - EmployeeID
 - LogDate
 - ActivityValue
 - PointsEarned
 - Status (Submitted/Verified)

4.5 Employee Assistance Program Management Module

Features

- Register EAP service types and manage confidential counselling session scheduling
- Track aggregate EAP utilisation statistics without revealing individual identities

Entities

- **EAPService**
 - ServiceID
 - ServiceName
 - Category
(MentalHealthCounselling/LegalAdvisory/FinancialCounselling/GriefSupport/ParentingAdvisory)
 - SessionsAllowedPerEmployee
 - Status (Active/Inactive)

- **EAPSession**
 - SessionID
 - EmployeeID
 - ServiceID
 - RequestedDate
 - SessionDate
 - CounsellorRef
 - Status (Requested/Scheduled/Completed/Cancelled)

4.6 Recognition & Rewards Management Module

Features

- Enable peer-to-peer recognition with category-based badges and point awards
- Manage milestone recognition for tenure, performance, and achievements with points redemption catalog

Entities

- **RecognitionAward**
 - AwardID
 - NominatorID
 - RecipientID
 - Category
(PeerRecognition/ManagerNomination/MilestoneAward/InnovationAward)
 - BadgeName
 - PointsAwarded
 - Message
 - AwardDate
 - Status (Awarded/Revoked)
- **RewardPoints**
 - PointsID
 - EmployeeID

- TotalEarned
- TotalRedeemed
- Balance
- LastUpdated
- **RedemptionCatalog**
 - ItemID
 - ItemName
 - Category (Voucher/Merchandise/Experience/Charity)
 - PointsRequired
 - AvailableQuantity
 - Status (Available/OutOfStock)

4.7 Benefits Analytics & Reporting Module

Features

- Enrolment coverage, plan participation, and premium cost dashboards by department and grade
- Wellness challenge completion rates, EAP utilisation statistics, and recognition activity trend reports

Entities

- **BenefitsReport**
 - ReportID
 - Scope (Department/Grade/Plan/Period)
 - Metrics (EnrolmentRate, PremiumCost, ClaimsSubmitted, WellnessParticipation, EAPUtilisation, RecognitionCount, PointsRedeemed)
 - GeneratedDate

4.8 Notifications & Alerts Module

Features

- Open enrolment window alerts, plan deadline reminders, and coverage confirmation notifications for employees

- Wellness challenge progress updates, EAP session confirmations, and recognition point credit alerts

Entities

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Enrolment/Wellness/EAP/Recognition/Benefits)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB
- **Production:** Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, EmployeeID, Name, Role, Email, Phone, GradeID, DepartmentID, Status)
- **AuditLog**(AuditID, UserID, Action, Module, Timestamp)
- **BenefitPlan**(PlanID, PlanName, PlanType, EligibilityGrade, EmployeeContribution, EmployerContribution, CoverageLimit, EffectiveDate, Status)
- **FlexBenefitBucket**(BucketID, PlanID, BucketName, AnnualAllowance, CarryForwardAllowed, Status)
- **EnrolmentWindow**(WindowID, PlanYear, OpenDate, CloseDate, EligibleGrades, Status)
- **BenefitEnrolment**(EnrolmentID, EmployeeID, PlanID, WindowID, CoverageOption, DependentsIncluded, EmployeeContributionAmount, EffectiveDate, Status)
- **Dependent**(DependentID, EmployeeID, Name, Relationship, DateOfBirth, Status)
- **WellnessProgram**(ProgramID, Name, Theme, StartDate, EndDate, PointsOnOffer, TargetParticipation, Status)

- **WellnessChallenge**(ChallengeID, ProgramID, ChallengeName, ActivityType, DailyTarget, Duration, PointsPerCompletion, Status)
- **ActivityLog**(LogID, ChallengeID, EmployeeID, LogDate, ActivityValue, PointsEarned, Status)
- **EAPService**(ServiceID, ServiceName, Category, SessionsAllowedPerEmployee, Status)
- **EAPSession**(SessionID, EmployeeID, ServiceID, RequestedDate, SessionDate, CounsellorRef, Status)
- **RecognitionAward**(AwardID, NominatorID, RecipientID, Category, BadgeName, PointsAwarded, Message, AwardDate, Status)
- **RewardPoints**(PointsID, EmployeeID, TotalEarned, TotalRedeemed, Balance, LastUpdated)
- **RedemptionCatalog**(ItemID, ItemName, Category, PointsRequired, AvailableQuantity, Status)
- **BenefitsReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Employee Benefits Portal:** My benefits summary, enrolment wizard, wellness challenge tracker, EAP booking, recognition wall
- **HR Benefits Admin Console:** Plan configurator, enrolment window manager, coverage report, dependent verification
- **Wellness Coordinator Dashboard:** Program builder, challenge activity feed, leaderboard, completion reward processor
- **Recognition Manager Console:** Award catalog, nomination review, milestone tracker, points balance view
- **Finance Dashboard:** Premium cost tracker, benefits budget utilisation, enrolment cost summary by plan
- **Admin Console:** Benefits catalog builder, eligibility tier setup, EAP service configuration, redemption catalog manager

8. Non-Functional Requirements

- **Performance:** Support 30,000+ concurrent employees during open enrolment windows and wellness challenge launches

- **Security:** EAP session data strictly anonymised in reports, dependent PII protection, RBAC, and full audit trails
- **Scalability:** Multi-country benefit plan support for global enterprises with region-specific plan configurations
- **Availability:** 99.9% uptime with enrolment and wellness modules prioritised during open enrolment peak periods
- **Maintainability:** Configurable benefit plan structures, flex bucket rules, wellness challenge templates, and reward catalogs
- **Observability:** Real-time enrolment progress dashboards, wellness participation metrics, and recognition activity monitoring

9. Assumptions & Constraints

- Phase 1 excludes integration with external insurance providers, payroll deduction systems, and wellness wearable APIs
- EAP counselling sessions are logged internally; integration with external EAP provider portals is deferred to Phase 2
- All notifications are in-app only; push notification and email digest integrations are out of Phase 1 scope
- Wellness activity logs are self-reported; wearable device and fitness app data sync is deferred
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling